DATA MANAGEMENT

LEARNING OBJECTIVES AND OUTCOMES

Explain the principles, frameworks, and importance of data management in companies

Learn basics of data preparation with Polyanalyst and perform exploratory data analysis with visualization tool PIX BI

1. Articulate the value of data management in a company, understand analytical methods
2. Design a basic governance framework (e.g., roles, data classification)
3. Create visualizations to identify patterns and outliers
4. Analyze a real-world dataset, resent findings in a business context, propose governance improvements based on analytical discoveries

LEARNING OBJECTIVES AND OUTCOMES

1 April: Introduction to Data Management & Data Analysis

8 April: Data Management Principles
Assessment type: quizz

15 April: Exploratory Data Analysis & Data Visualization
Assessment type:quizz

May-June: Practice on Polyanalyst and PIX BI tool
Assessment type: Final project (in groups)

GROUP WORK AGREEMENTS

1. Be active, but no one is forced to speak.
2. The "Swedish table" rule—take what you find valuable and relevant.
3. Let others speak - take turns talking.
4. Consider the interests of other participants.
5. Learn here and now = "No mistakes, only feedback."
6. Return from breaks on time.
7. Ask questions = "There are no stupid questions - only feedback".
8. Relax and enjoy you time.

WHAT I EXPECT FROM THE TRAINING

Introduce yourself:

- Your name
- Your group
- **What do you know about Data Management and BI?**



DATA MANAGEMENT BASICS

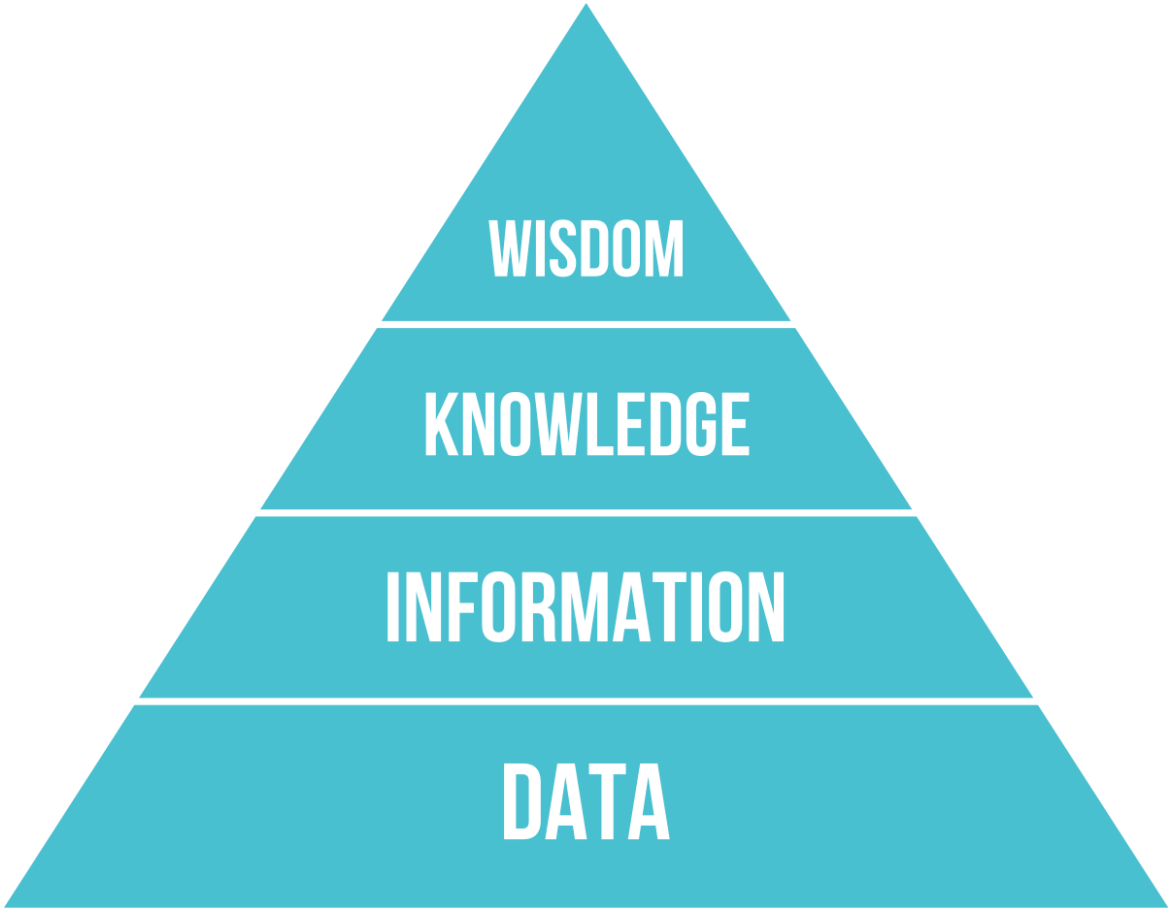
DIKW

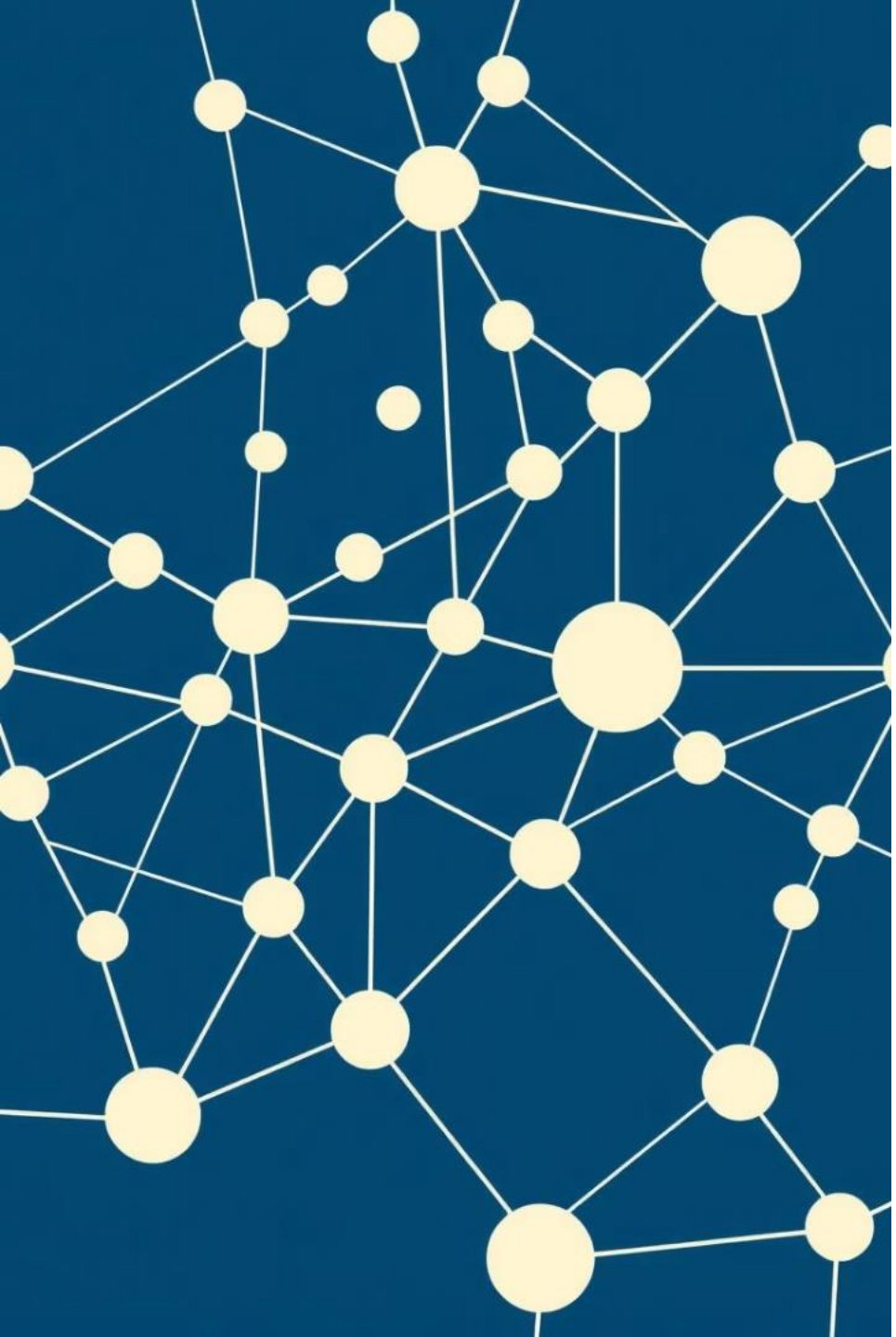
W — **Wisdom —**
Application of knowledge to make decisions

K — **Knowledge —**
Insights derived from information

I — **Information —**
Processed data with context

D — **Data —**
Raw facts and figures





DATA AS A CORPORATE ASSET

Not Individual

Data does not belong to individual employees but is used for analytics across different company departments

Daily Operations

Essential for organizations to ensure daily operational activities

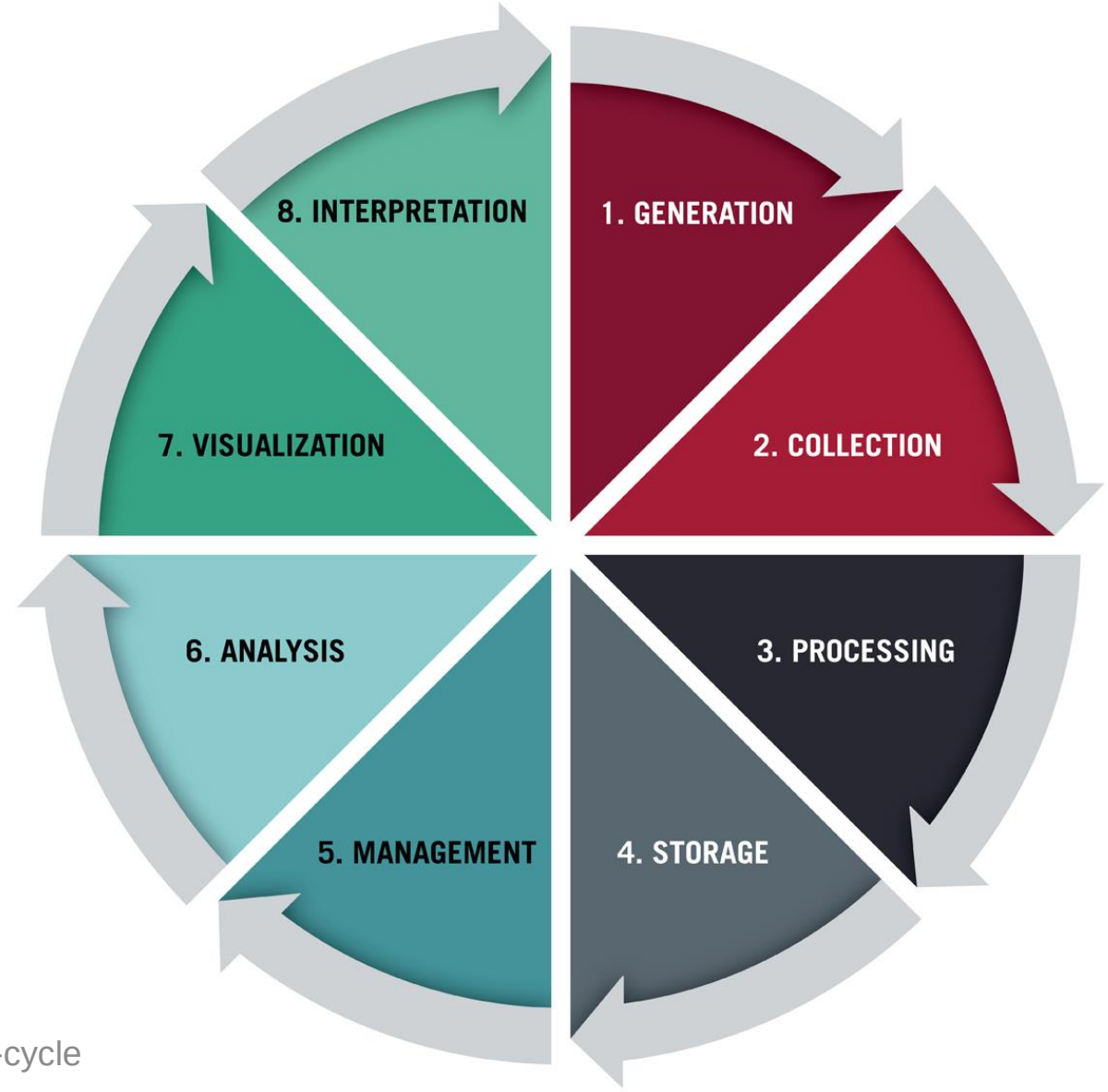
Improve Efficiency

Data provides opportunities to improve efficiency and optimize company operations

Employee Contribution

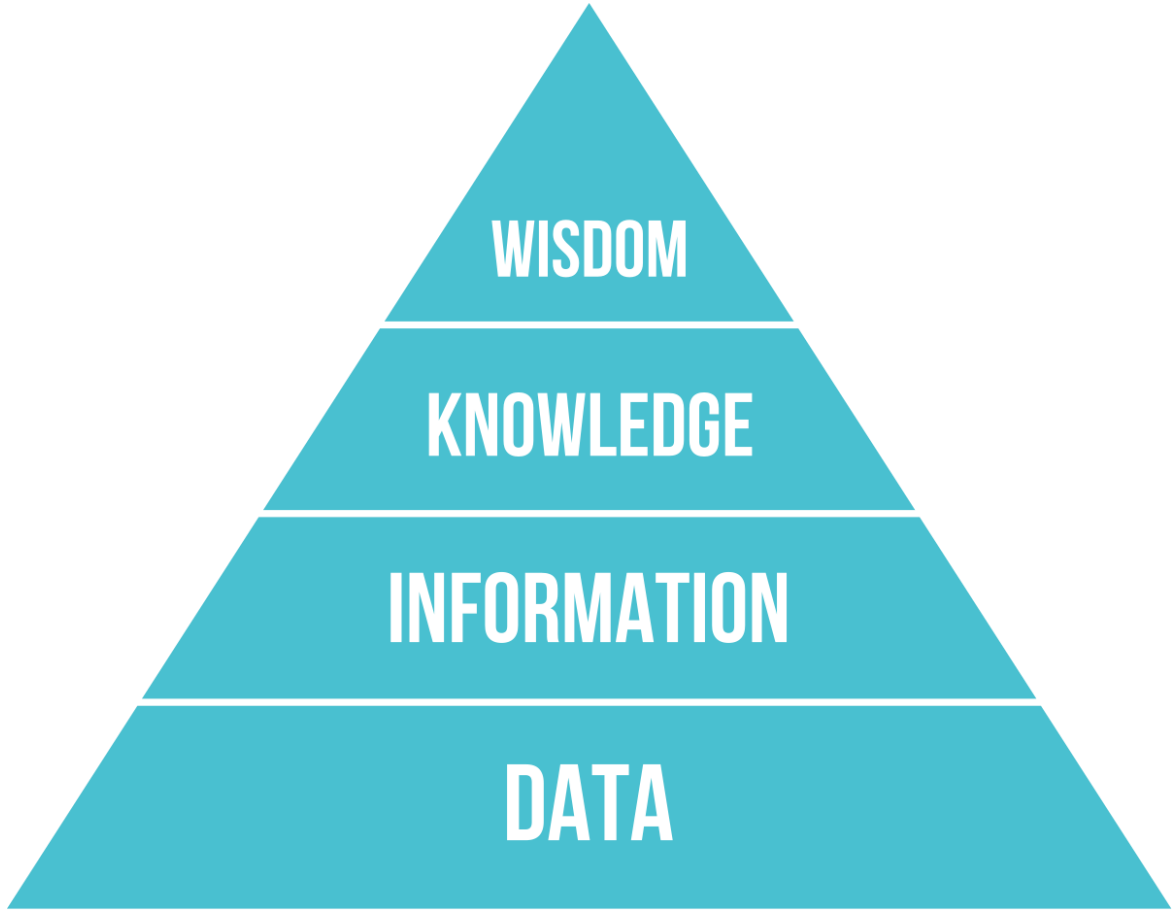
Every employee generates data in their work, which others transform into valuable information.

DATA LIFECYCLE



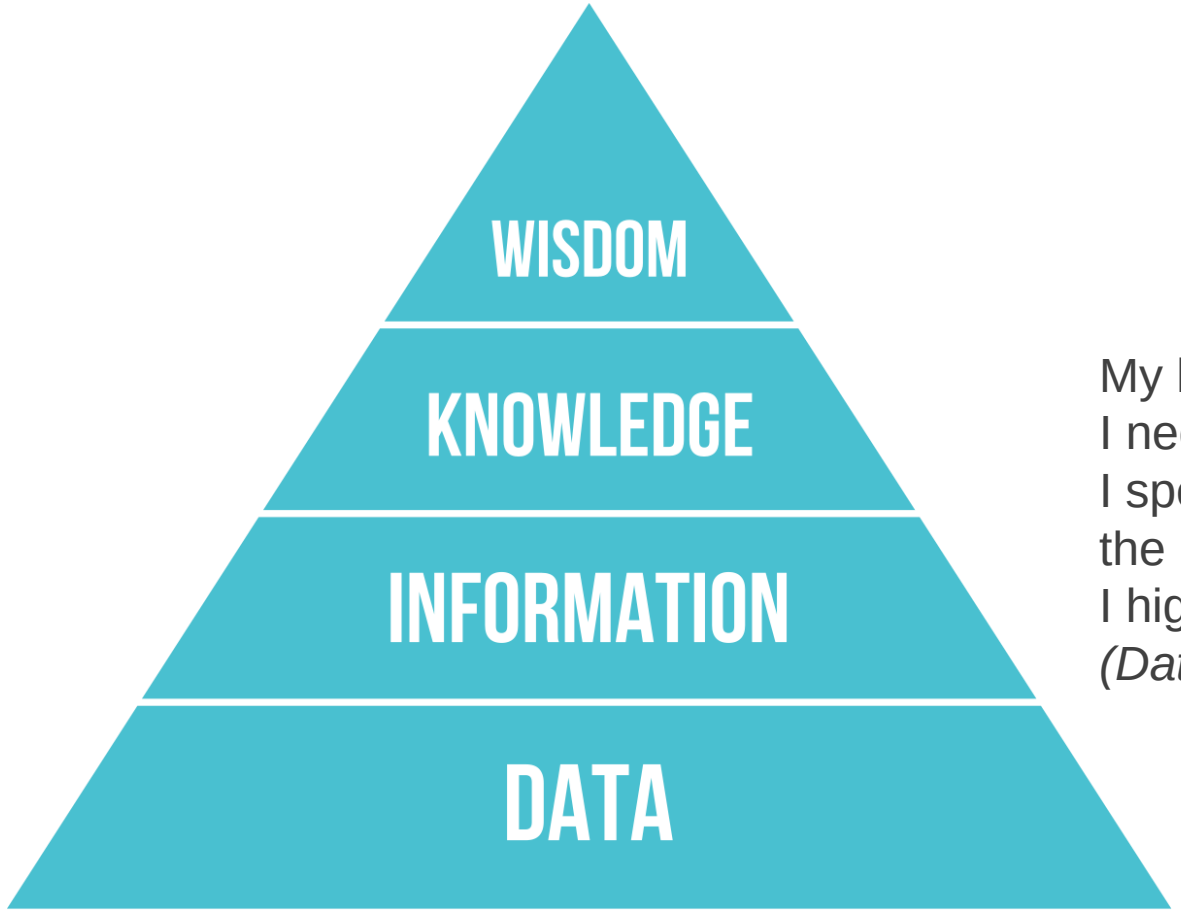
https://online.hbs.edu/blog/post/data-life-cycle

PREPARING FOR AN EXAM



72%
Chapter 3, Pages 45-50
10:00 AM-12:00 PM, Library
5 key terms highlighted

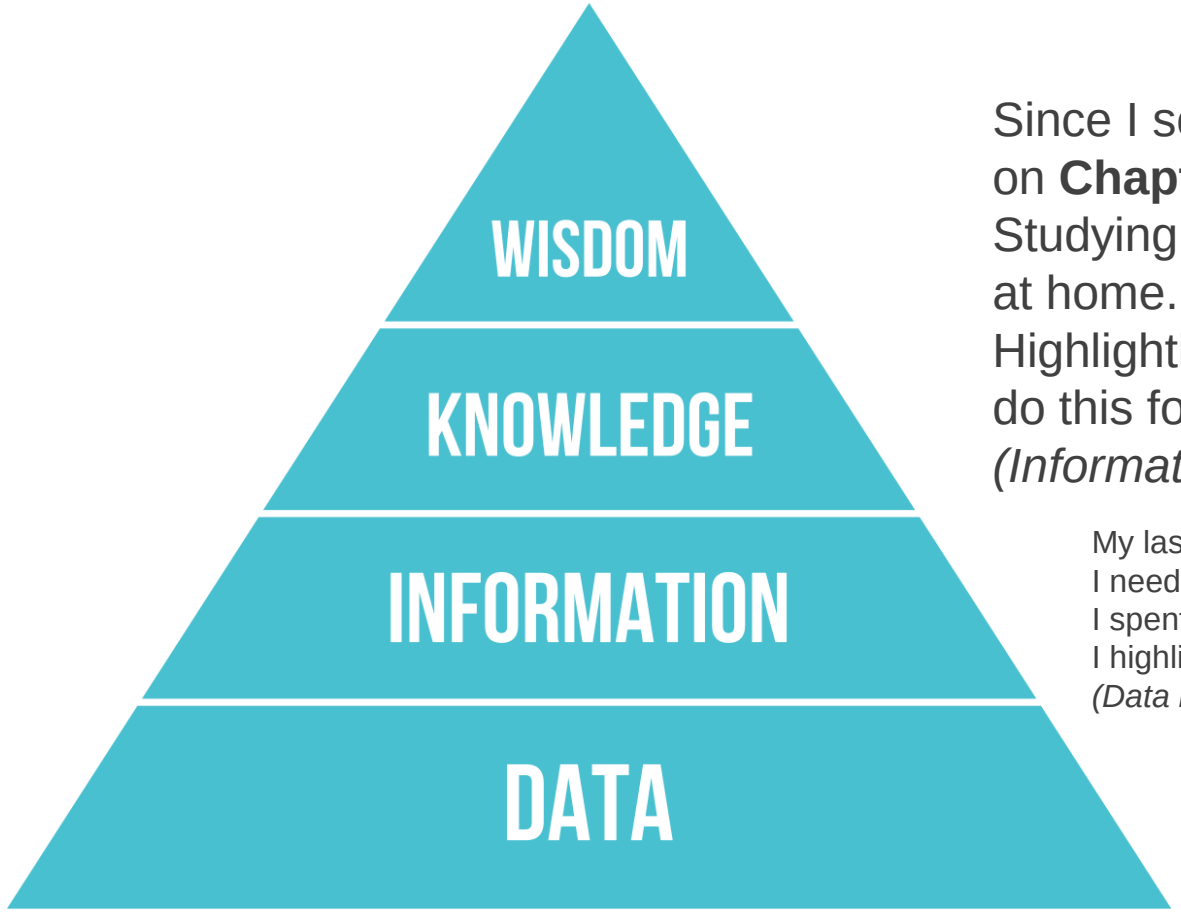
PREPARING FOR AN EXAM



My last quiz score was **72%**.
I need to study **Chapter 3 (Pages 45-50)** for the exam.
I spent **2 hours (10:00 AM - 12:00 PM)** studying at the **library**.
I highlighted **5 key terms** in my notes.
(Data is processed into meaningful facts with context.)

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PREPARING FOR AN EXAM

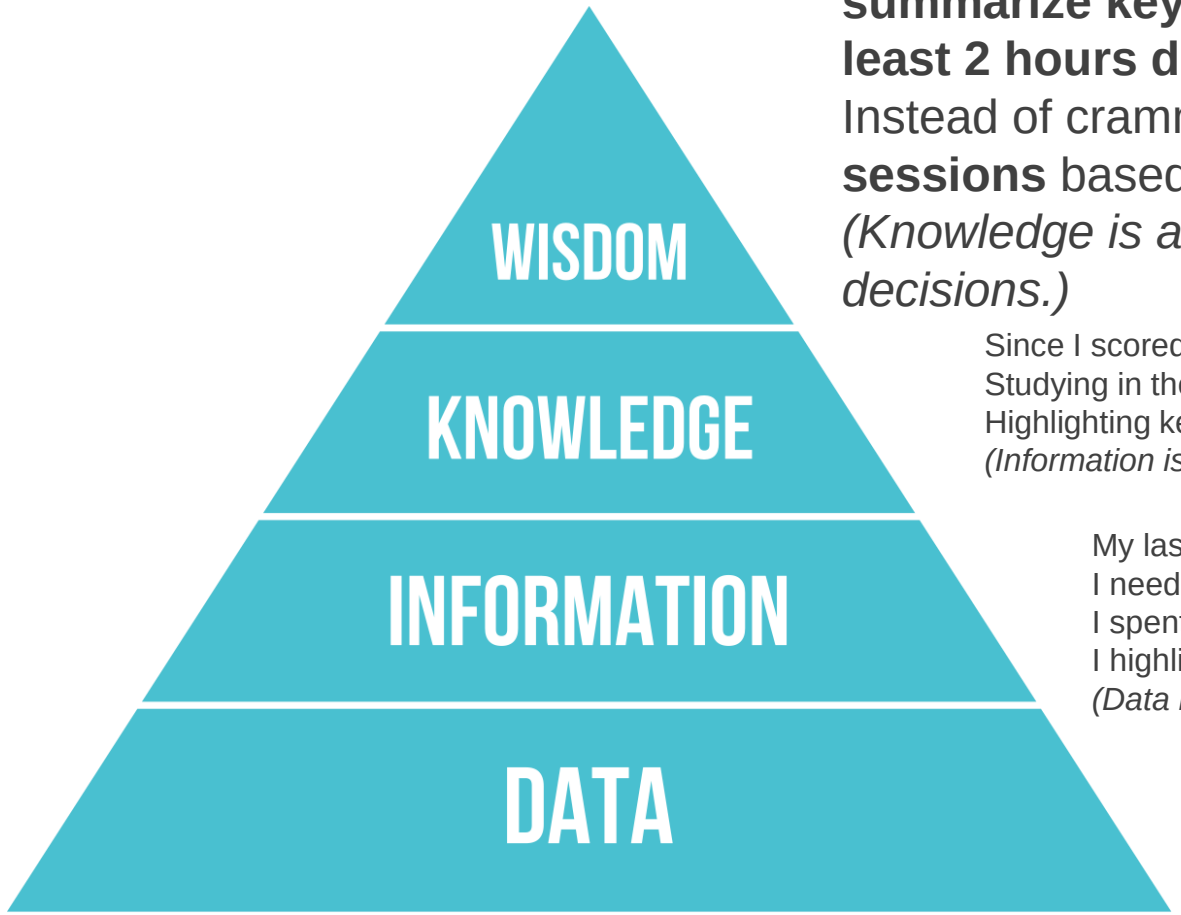


Since I scored **72%** on the quiz, I need to focus more on **Chapter 3** where I lost marks.
Studying in the **library** helps me concentrate better than at home.
Highlighting key terms improves my retention, so I should do this for all chapters.
(Information is analyzed to form strategies and insights.)

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PREPARING FOR AN EXAM



To improve my grade, I will **revise Chapter 3 first, summarize key terms, and study in the library for at least 2 hours daily**.
Instead of cramming, I will **space out my study sessions** based on what worked last time.
(Knowledge is applied with experience to make effective decisions.)

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PREPARING FOR AN EXAM



I will adjust my study plan to focus on weak areas and use proven techniques.

I need to study Chapter 3 more because of my low score.

I scored 72% on the quiz.

72%.

DATA-DRIVEN – MANAGEMENT BASED ON DATA & ANALYTICS

Key feature: decisions based on data and analytics, not intuition

PRINCIPLES:

Investment in data infrastructure

Data must be extracted, stored, analyzed, interpreted, and visualized. All of this requires attention, time, and money.

Ability to analyze and interpret data

An important part of working with data is analyzing it and building hypotheses. This requires special knowledge and experience.

Trust in data quality

The data must be of high quality (accurate and clean) — then it can be trusted and interpreted correctly.

Data-driven decision-making

Before making any important decision, you need to collect and analyze the data.

DATADRIVEN COMPANY

Industrial Company

Digital (Data-Centric) Company

PEOPLE-BASED

PROCESS-BASED

DIGITAL TRANSFORMATION

DATA-BASED OPERATIONS

MODEL-BASED OPERATIONS

PEOPLE

PROCESS

DATA

MODELS

→ Effectiveness of employee skills and business processes no longer creates additional value but is the foundation of business viability

→ **Competitive advantage now lies in improving data efficiency**

→ Transitioning to a data-centric paradigm requires implementing all basic aspects of data management in the company.



Gazprom Neft's Data Management Strategy

Target Focus

Ensure data accessibility and quality.

Create new value from insights.

Means of Achieving Goal

Implement policies and standards.

Improve data culture and competencies.

Target Data Characteristics

Accessible, high-quality data.

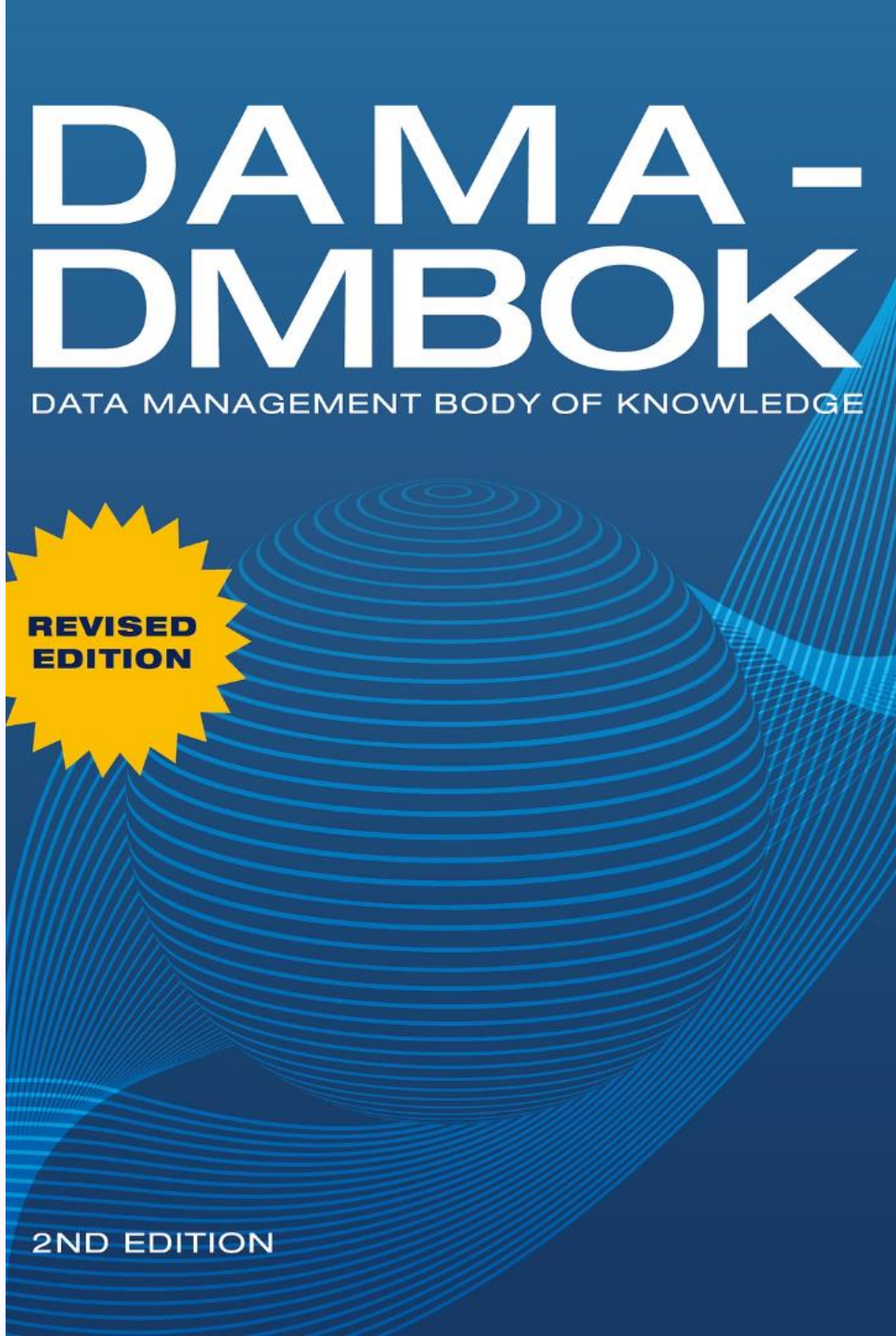
Relevant data with analytical potential.

Made with Gamma

DAMA DMBOK: Data Management Body of Knowledge

DAMA DMBOK provides a comprehensive framework. It covers key aspects of data management. This includes data governance and data quality. Effective data strategy is also part of it.

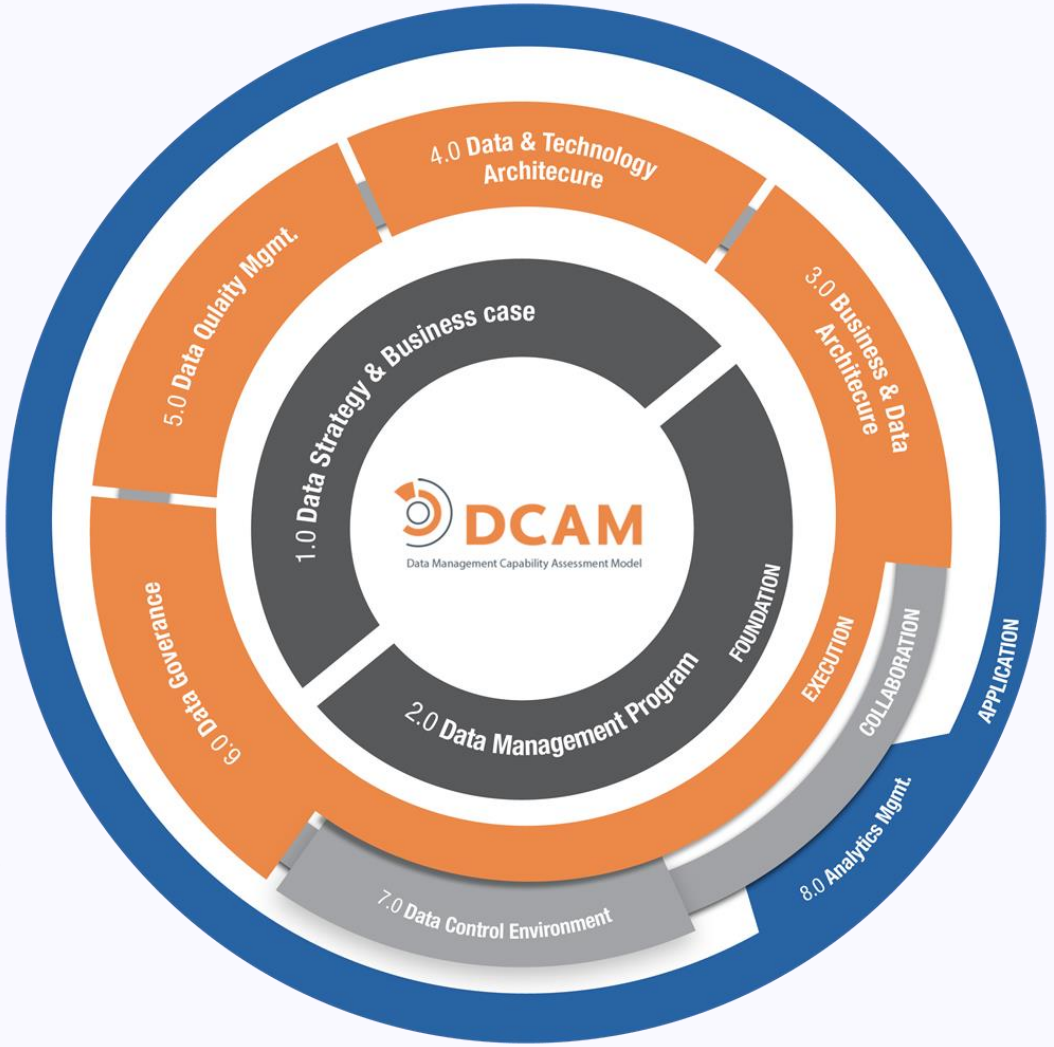
Use DAMA DMBOK to improve your organization’s data practices. This will lead to better decision-making.



DCAM Framework: Data Capability Assessment Model

The DCAM framework assesses data management capabilities. It identifies areas for improvement. This model promotes consistent data practices. It also helps achieve business goals.

Implement DCAM to mature your data capabilities. Better data management supports strategic objectives.



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BREAK

BASICS OF DATA ANALYSIS

Analytics Value Chain: From Data to Strategy

Data

1

Data Reports

Reflect current business state.

2

Data Analysis

Identify key problems and trends.

3

Management Decisions

Informed by data analysis.

4

Value

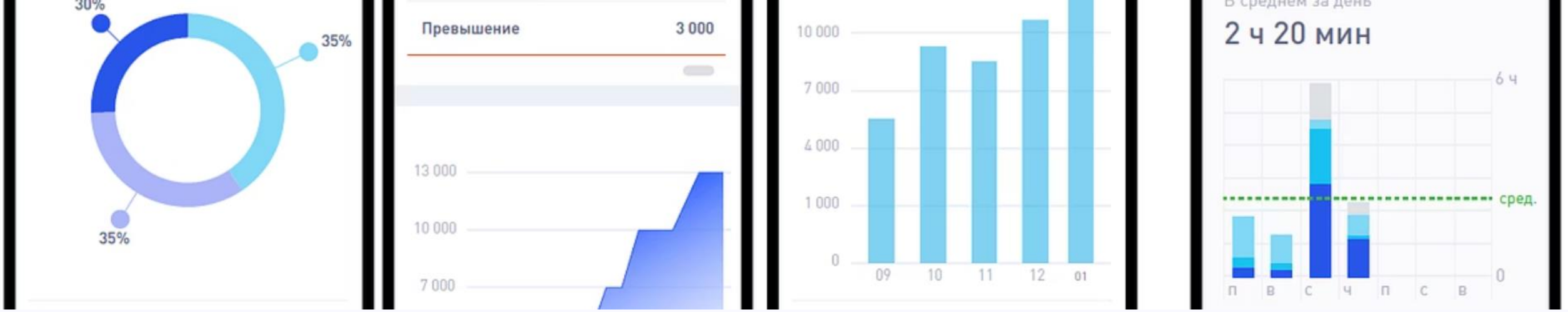
Data and analysis results can influence company strategy and development.



DATA ANALYSIS

DATA ANALYSIS – process of researching, filtering, transforming, and modeling data to extract insights and make decisions





Applying Data Analysis Techniques

Use data analysis to fix your life. For instance, analysis of finance in a mobile bank, that reflect statistics on expenses. Based on the analysis of such data, it is possible to make a decision to reduce costs on something less needed.

Gain competitive advantage by leveraging data.

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GROUP WORK

DATA ANALYSIS IN DAY TO DAY LIFE

Make a list of 5 examples of data analysis in everyday life:

1.

2.

3.

4.

5.

TIME
10 MINUTES



Reporting vs. Data Analysis

Reporting: Organizing data into summaries to track business performance

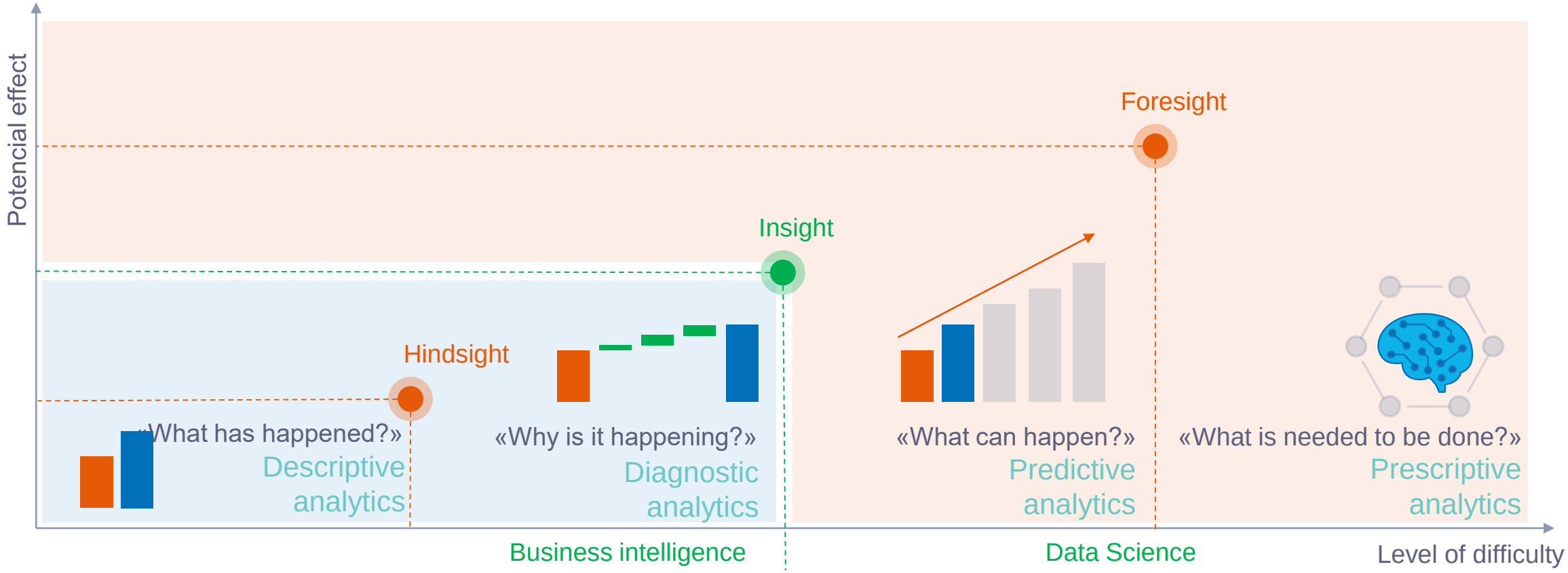


Analysis: Transforming data into insights that drive decisions through people, processes, and technology



Reporting	Analysis
Shows what happened (retrospective)	Explains why it happened
Raises questions	Provides recommendations
Data → Information (no context)	Data + Information → Insights (with context)

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Profit&Loss: collection and aggregation of actual indicators for the completed period

Cash flow: collection of veils from bank transactions for consolidated items of management accounting

What if analysis: identification of factors that could be changed to achieve the planned results

ABC-analysis: identifying the most important elements

Oil price forecast: forecasting prices based on production plans, market demand, currency prices, sanctions, and other factors

Route in navigator: selecting the most efficient routes according to the time of day, traffic congestion, road events, speed limits, and other factors.

Basic Tools for Working with Data

Excel: Uses and Limitations:

Uses:

- Simple analytical tasks (pivot tables, basic formulas, VLOOKUP).
- Manual data entry.
- User-friendly for most.
- Can serve as a data source for BI.

Limitations:

- Complex formulas and transformations are slow.
- High risk of errors due to manual input.
- Row limit (1 million).
- Limited automation and visualization options.



Business Intelligence (BI) vs. Data Science

BI Answers:

What happened? Why is it happening?

Data visualization, transformation, and reporting automation.

Works with structured data.

Data Science Answers:

What will happen next? What should we do?

Uncovers hidden patterns, uses historical data for predictions.

Applies complex models (forecasting, classification, clustering).

Works with structured and unstructured data.



















APROACHES TO BI IMPLEMENTATION AT GAZPROM NEFT

Business Intelligence (BI)

End-to-end development of analytical applications

Self Service BI

Business users independently developing applications for routine data processing and visualization needs



Business user

Business requirements

Turnkey analytical applications



Data Analytics Team

Data request

Curated data, tools and expertise



Business user

Self-built applications





	Business Intelligence (BI)	Self-Service BI
Tasks	Automation of reporting.	On-demand analytics.
Data	Data from source systems.	Local user files.
BI developers	BI developers needed.	Business users create apps.
Tools	Luxms BI	PIX BI делает умнее

DATA ANALYTICS SKILLS

Data literacy and comprehension

Business context understanding
(Grasping the real-world business implications of data)

Data format/type identification
(Recognizing and classifying data structures - JSON, CSV, etc.)

Data quality standards awareness
(Knowing dimensions of data quality: completeness, accuracy, timeliness)

Data analytics lifecycle
(End-to-end process from collection to insights to archiving)

Data sensitivity and preservation
(Handling confidential data and ensuring its long-term integrity)

Data processing

Data transformations (cleansing, formatting, etc.)

Computations

Data testing/validation

Implementation

Evidence-Based Reasoning" captures the analytical rigor behind arguments

Data Storytelling" remains in English as an industry-standard term

Exploratory Data Analysis (EDA)

Visual Literacy / Data Visualization Comprehension
(Ability to correctly interpret charts, graphs, and other data visualizations)

Insight Generation & Analytical Reasoning
(Competency to derive meaningful conclusions and actionable insights from data)





Objective: use basic data governance and BI tools to fix messy sales data and create useful reports.

Scenario: The Problem with Sales Reports

Company's sales team has trouble tracking fuel sales because:

- Sales data comes from 5 different regions, each using different Excel files.
- Some files have missing customer names or wrong prices.
- Managers can't see real-time sales trends.

Your Task: Help the sales team by:

- **Cleaning up the data** (management).
- **Creating a simple dashboard** (BI tool).

Step 1: Fix the Data (20 min)

Question: How can we make sure sales data is correct and easy to use?

Ideas:

Make a rule: "All regions must use the same Excel template."
Assign someone to check for missing/wrong data every week.
Save files in one shared folder (no more emailing Excel files!).

Deliverable: Write down **2 simple rules** for better sales data.

Step 2: Build a Dashboard (20 min)

Tool: Imagine using **Power BI** or **PIX BI**.

Question: What should the dashboard show to help managers?

Ideas: A bar chart of "Monthly Sales by Region."

Step 3: Present your results